Exemple

$$U(z) = \sum_{k=0}^{+\infty} u_k z^{-k} = \sum_{j=0}^{+\infty} u_{2j} z^{-2j} + \underbrace{u_{2j+1} z^{-2j-1}}_{=0}$$

$u_k = 0$ si k impair.

$u_k = 1$ si k pair

$$= \sum_{j=0}^{+\infty} (u_{2j}) z^{-2j}$$

$$= \sum_{j=0}^{+\infty} \left(z^{-2}\right)^j = \frac{1}{1 - z^{-2}} = \frac{z^2}{z^2 - 1}$$

$$= \frac{z^2}{(z+1)(z-1)}$$

Exemple : intégrateur numérique

$$y_{k+1} = y_k + u_k T_e \quad / \quad y_0 \text{ est donné. } \forall k \geq 0$$

* $q y_k - y_k = T_e u_k \iff (q-1) y_k = T_e u_k$

Si $y_0 = 0$, après $\mathcal{Z}\{\ \} \longrightarrow (z-1) Y(z) = T_e U(z) \iff \boxed{Y(z) = \frac{T_e}{z-1} U(z)}$

* $y_k = y_{k-1} + u_{k-1} T_e$

$$y_k = q^{-1} y_k + q^{-1} T_e u_k \iff (1 - q^{-1}) y_k = q^{-1} T_e u_k$$

Si $y_0 = 0$, après $\mathcal{Z}\{\ \} \longrightarrow (1 - z^{-1}) Y(z) = T_e z^{-1} U(z)$

$$\iff \boxed{Y(z) = \frac{T_e z^{-1}}{1 - z^{-1}} U(z)}$$

$$= \frac{T_e}{z-1} U(z)$$

$$S_k = \sum_{i=0}^{k} u_i = \left(\sum_{i=0}^{k-1} u_i\right) + u_k \qquad \xrightarrow{\mathcal{Z}\{\}}$$

$$S(z) = z^{-1} S(z) + U(z)$$

$$\Uparrow$$

$$(1 - z^{-1}) S(z) = U(z)$$

$$S(z) = \frac{1}{1 - z^{-1}} U(z)$$

$$= \frac{z}{z - 1} U(z)$$

$u_k, k \in \mathbb{N}$

$$\boxed{S_k = S_{k-1} + u_k}$$

Exemple

$$\frac{W(z)}{z} = \frac{z+3}{(z-1)(z+2)z} = \frac{C_0}{z} + \frac{C_1}{z-1} + \frac{C_2}{z+2}$$

$$C_0 = -3/2 \qquad C_1 = 4/3 \qquad C_2 = 1/6$$

$$W(z) = C_0 + \frac{C_1 z}{z-1} + \frac{C_2 z}{z+2}$$

Z^{-1}

$$w_k = C_0 \delta_k + C_1 1_k + C_2 (-2)^k$$